

YAKUTAT, AK, USA

WMO: 703610

Lat: 59.512N

Lon: 139.671W

Elev: 10

StdP: 101.20

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25339

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-13.2	-22.0	0.5	-11.5	-19.2	0.7	-9.3	11.4	3.1	10.2	2.7	0.4	100	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.6	20.3	13.2	18.2	13.1	16.8	12.9	15.0	17.8	14.3	16.7	13.8	15.8	3.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	9.9	15.5	13.3	9.5	14.9	12.7	9.2	14.4	41.7	17.7	40.0	16.9	38.4	15.8	19.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.4	6.0	-18.4	24.5	2.9	3.3	-20.5	26.9	-22.3	28.9	-23.9	30.7	-26.0	33.2		
(5)			-18.7	16.0	2.9	1.4	-20.8	17.0	-22.5	17.8	-24.1	18.6	-26.1	19.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	-1.9	-1.0	-0.2	3.4	7.3	10.8	12.8	12.5	9.6	5.4	0.7	-1.0	(6)
(7)		DBStd	6.28	5.10	4.18	3.78	2.62	2.42	2.05	1.49	1.64	2.27	2.78	4.07	4.42	(7)
(8)		HDD10.0	2089	368	307	317	197	89	12	1	2	33	143	280	341	(8)
(9)		HDD18.3	4901	626	541	576	447	341	224	172	181	262	401	530	599	(9)
(10)		CDD10.0	231	0	0	0	0	6	37	87	79	21	1	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	19	0	0	0	0	0	9	4	6	0	0	0	0	(12)
(13)		CDH26.7	3	0	0	0	0	0	1	1	1	0	0	0	0	(13)
(14)	Wind	WSAvg	2.3	2.3	2.3	2.5	2.3	2.4	2.1	2.0	1.9	2.2	2.4	2.3	2.5	(14)
(15)	Precipitation	PrecAvg	3703	313	267	256	226	221	160	203	321	490	525	350	363	(15)
(16)		PrecMax	6356	808	816	947	486	481	466	474	718	1228	1240	1085	894	(16)
(17)		PrecMin	2601	60	5	64	49	41	32	47	61	62	76	46	96	(17)
(18)		PrecStd	781	166	153	154	107	110	103	99	173	194	201	176	186	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	7.7	10.8	17.2	21.0	24.7	22.2	23.2	20.6	13.3	10.0	9.1	(19)
(20)			MCWB	6.3	5.0	5.9	9.0	11.0	14.1	14.7	14.3	12.5	8.3	8.4	6.8	(20)
(21)		2%	DB	6.2	6.3	8.0	12.8	17.5	18.9	18.8	19.5	17.0	11.9	8.5	6.7	(21)
(22)			MCWB	5.2	4.7	4.6	6.6	9.8	13.1	14.3	14.0	11.7	9.4	7.2	5.5	(22)
(23)		5%	DB	5.2	5.3	6.4	10.1	14.3	16.3	17.2	17.6	15.1	10.9	7.2	5.6	(23)
(24)			MCWB	4.3	4.2	4.0	5.7	9.2	12.3	13.8	13.8	11.7	9.0	5.9	4.6	(24)
(25)		10%	DB	4.2	4.3	5.0	8.3	12.2	14.9	16.1	16.2	13.9	9.9	6.1	4.4	(25)
(26)			MCWB	3.4	3.3	3.2	5.4	8.8	11.7	13.4	13.5	11.7	8.2	4.9	3.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	6.3	7.5	9.8	11.9	15.2	15.8	16.2	14.3	11.4	9.1	7.5	(27)
(28)			MCDWB	7.2	6.8	9.3	16.3	18.1	20.9	19.1	19.6	15.8	11.8	9.6	8.7	(28)
(29)		2%	WB	5.5	5.2	5.8	7.7	10.7	13.8	14.9	15.1	13.4	10.4	7.7	5.8	(29)
(30)			MCDWB	6.0	5.9	7.2	11.3	14.8	17.4	17.6	17.5	15.0	11.0	8.4	6.6	(30)
(31)		5%	WB	4.5	4.4	4.5	6.6	10.0	12.9	14.2	14.5	12.7	9.4	6.3	4.7	(31)
(32)			MCDWB	5.0	5.1	5.7	9.0	13.5	15.8	16.5	16.7	14.3	10.3	7.0	5.4	(32)
(33)		10%	WB	3.5	3.5	3.5	5.7	9.2	12.0	13.7	13.8	11.9	8.3	5.1	3.7	(33)
(34)			MCDWB	4.0	4.2	4.7	7.9	11.8	14.1	15.7	15.8	13.4	9.4	6.0	4.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.6	6.9	7.9	8.1	7.7	6.5	5.6	6.8	7.4	6.9	6.1	5.3	(35)
(36)			MCDBR	4.6	6.5	9.4	12.9	14.2	11.2	9.3	11.3	11.5	8.5	6.3	5.3	(36)
(37)			MCWBR	4.0	5.0	6.6	7.8	7.9	6.2	5.5	6.9	7.4	6.2	5.3	4.4	(37)
(38)		5% WB	MCDBR	4.3	5.1	7.1	10.1	10.6	9.3	7.5	8.6	7.8	5.6	5.4	5.0	(38)
(39)			MCWBR	4.0	4.3	5.4	6.5	6.3	5.4	4.6	5.7	5.6	4.8	5.0	4.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.283	0.298	0.320	0.363	0.386	0.387	0.373	0.377	0.353	0.323	0.280	0.245	(40)	
(41)		taud	2.130	2.297	2.273	2.194	2.222	2.305	2.391	2.383	2.412	2.346	2.151	2.058	(41)	
(42)		Ebn at Noon	489	717	812	832	839	845	850	815	773	672	485	389	(42)	
(43)		Edh at Noon	46	67	92	119	125	117	106	99	82	63	44	33	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.36	1.12	2.56	4.21	5.30	5.52	4.96	3.90	2.53	1.28	0.51	0.21	(44)	
(45)		RadStd	0.04	0.15	0.24	0.30	0.40	0.25	0.25	0.37	0.32	0.15	0.07	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.56	N/A	N/A	N/A	N/A	N/A	N/A	-223	N/A	N/A
(47) Regional (2 neighbors)		N/A	+1.47	+1.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page